Jinay Patel

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Education

University of Waterloo
Bachelor of Computer Engineering — *Expected Graduation: Jun 2031*

Skills

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years)
Technologies: PyTorch, LangChain, Mastra, CMake, vcpkg, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity, Linux

Experience

Lead Software Engineer — October 2023 — June 2026
Seaquam Robotics — *Delta, BC*

- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by 50%
- Created a custom React and Tauri desktop CMS app, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server **response speed by 150%**

Embedded Controls Software Engineer — October 2021 — June 2023
Seaquam Robotics — *Delta, BC*

- Implemented C++ motion algorithms, securing a **100% win rate** during autonomous periods across 15+ matches
- Designed an interpolated lookup table using a monotonic cubic spline to tune PID constants in real time, increasing movement accuracy by 200% with **error tolerances under 1 inch**

Software Engineering Tutor — September 2025 — March 2026
Code Ninjas — *Surrey, BC*

- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving the local branch's search visibility

Freelance Systems Administrator / IT Consultant
Convergence Media — *Remote*

- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a **Tailscale tunnel** to securely expose the Nextcloud server for remote video calls and file sharing

Projects

Content Management System & Website Backend | Lead Software Engineer — GitHub

- Built a backend to aggregate data from an external API and a custom content management system that let our coach update data on the robotics website without touching code, reducing reliance on developer support for routine content changes
- Built the backend using **Deno** and created the desktop app for content management system using **Tauri** and **React**

Telemetry Data Visualization | Embedded Controls Software Engineer — GitHub

- Streamed CSV-formatted odometry, PID error data, and logs over a **websocket** to a **real-time Svelte dashboard**
- Used Chart.js to render live PID and error graphs and the Canvas API to draw the robot's live position on a 2D field

AI Complaint Categorization System | Hackathon Project — GitHub

- Built a pipeline that analyzed text, audio, and image-based complaints, categorized them, and stored the results in a **vector database** for a **RAG pipeline**; named a **finalist at the Headstarter Fellowship Hackathon**
- Leveraged OCR Space, Gemini, Hugging Face, LangChain, and Supabase (for the vector database) to power the pipeline

Awards

Vex Robotics World Champion, RECF & Vex Robotics — April 2026
Top overall team among 20,000+ worldwide, recognized for excellence across robot design, programming, and competitive performance

Excellence Award @ Provincial Championship, Vex Robotics — February 2026
Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks — July 2025
Awarded for developing Posturai, a real-time posture correction app

Senior Computer Programming Award, Seaquam Secondary School — June 2026

Euclid Math Contest Distinction, Waterloo CEMC — May 2026
Top 25% in Canada for the Euclid math contest